
PRACTICE FINAL EXAM

NAME: _____

ID N°: _____

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- When solving a problem, if you are going to use an alternative method to the ones discussed in class (for instance, inverting a matrix via adjugates or using Cramer's rule to solve a linear system), you are required to include a short explanation on how this alternative method works.
- The use of calculators of any type is not allowed.

Good Luck!

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Problem 1. Define the following concepts:

- (a) Let A be an $n \times n$ matrix. State precisely what it means for A to be **invertible**, and what it means for an $n \times n$ matrix B to be an **inverse** of A .
- (b) Let V be a vector space and $B = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ a set of vectors in V . Define precisely what it means for B to be a **basis** of V , explaining the concepts that are involved in the definition of basis.

- (c) Let A be an $m \times n$ matrix. Define precisely the **column space** and the **image space** of A .
- (d) Let A be an $n \times n$ matrix and $\lambda \in \mathbb{R}$ a number. Define precisely what it means for λ to be an **eigenvalue** of A .
- (e) Let V be a vector space and $U \subseteq V$ a subset. Define precisely what it means for U to be a **subspace** of V .

Problem 2. Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ -6 & -8 & -10 & -12 \end{bmatrix}$$

and the vector $\mathbf{b} = (0, 4, -4)$.

- (a) Compute the **reduced** row-echelon form of A and use it to give bases for $\text{row } A$ and $\text{col } A$.

- (b) Using part (a), compute a set of basic solutions to the system $A\mathbf{x} = \mathbf{0}$.
- (c) The vector $\mathbf{z} = (1, 1, -1, 0)$ is a solution to the system $A\mathbf{x} = \mathbf{b}$. Use this fact and part (b) to compute the general solution to the system $A\mathbf{x} = \mathbf{b}$. Write your answer as the sum of a particular solution to the system together with the general solution of the associated homogeneous system.

Problem 3. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}.$$

- (a) Compute the eigenvalues of A , and for each eigenvalue of A compute its corresponding eigenvectors.

- (b) Is A diagonalizable? If so, find a diagonal matrix D and an invertible matrix P such that $D = P^{-1}AP$. If not, explain why.

Problem 4. Calculate the determinant of

$$D = \begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ -1 & 3 & 1 & 4 & 0 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & -1 & 0 & 2 \\ 0 & 0 & 3 & 0 & 1 \end{bmatrix}.$$

Problem 5. Let $\mathbf{M}_{22}$ be the space of all 2×2 matrices and consider

$$B = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}.$$

We define the subset

$$V = \{A \in \mathbf{M}_{22} \mid \operatorname{tr}(AB) = 0\}.$$

(a) Show that V is a vector subspace of $\mathbf{M}_{22}$.

(**Hint.** It may be useful to recall that the trace tr is a linear transformation).

(b) Find a basis of V . Is V isomorphic to $\mathbb{R}^2$?

Problem 6. For each natural number n , we denote by $\mathbf{P}_n$ the vector space of all polynomials of degree at most n , including the zero polynomial. We define a linear transformation $T: \mathbf{P}_2 \rightarrow \mathbf{P}_2$ by the equation

$$T(p(x)) = xp'(x) - 2x^2p(-3), \quad \text{for all } p(x) \in \mathbf{P}_2.$$

- (a) If $B = \{1, x, x^2\}$ denotes the standard basis of $\mathbf{P}_2$, find the matrix $[T]_{BB}$ of T relative to B and B .

- (b) Let $q(x) = -2x + 1$. Find the coordinate vector $[q(x)]_B$ of $q(x)$ with respect to B .

- (c) Use the matrix $[T]_{BB}$ to compute $[T(q(x))]_B$ and $T(q(x))$.

Problem 7. Decide whether each of the statements below is true or false. Completely fill the circle that is to the left of your choice. You do not need to justify your choices.

(a) If A is a 3×3 matrix and A is diagonalizable, then A has three different eigenvalues.

☐ True

☐ False

(b) The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $T(x, y) = |x|$ is linear.

☐ True

☐ False

(c) If a consistent system of linear equations has more variables than equations, then it has infinitely many solutions.

☐ True

☐ False

(d) The set $\{1 + x, \sin x, \cos x\}$ is linearly independent in $\mathbf{F}[0, 1]$.

☐ True

☐ False

(e) A linear transformation $T: V \rightarrow W$ is one-to-one if and only if $\text{im } T = W$.

☐ True

☐ False

(f) If V and W are finite dimensional spaces with $\dim V = \dim W$, then V and W are isomorphic.

☐ True

☐ False